

Where we left off...

- Every SISO-LTI-system can be written as a fraction of two polynomials in frequency domain.

Time domain

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

Laplace-Transform

$$G(s) = C(sI - A)^{-1}B + D$$

Frequency domain

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d$$

- The variable s is a "complex frequency".

$s = \text{_____}$, where

• $\text{Re}(s) = \sigma$: _____ part

• $\text{Im}(s) = \omega$: _____ part

$\Rightarrow$ In this course, most of the time we have
 _____ (pure sinusoids).

• We saw, that most signals can be written as

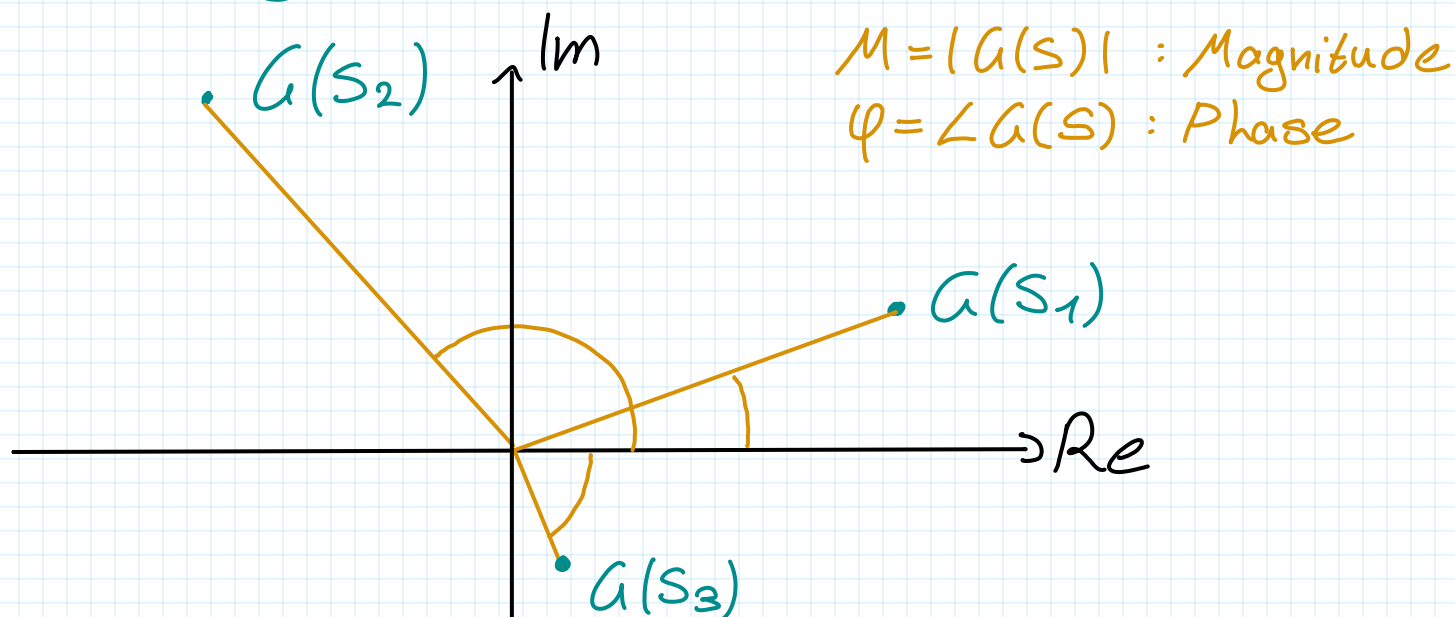
$$u(t) = \text{_____} \Rightarrow \text{Ex: } \cos(\omega t) = \text{_____}$$

• Since we had that $u \rightarrow \boxed{G(s)} \rightarrow y$, we find that

$y_{ss}(t) = \text{_____}$, but if we evaluate this for some $s = \sigma + j\omega$, we get weird complex values...

$$\Rightarrow y_{ss}(t) = \text{_____}$$

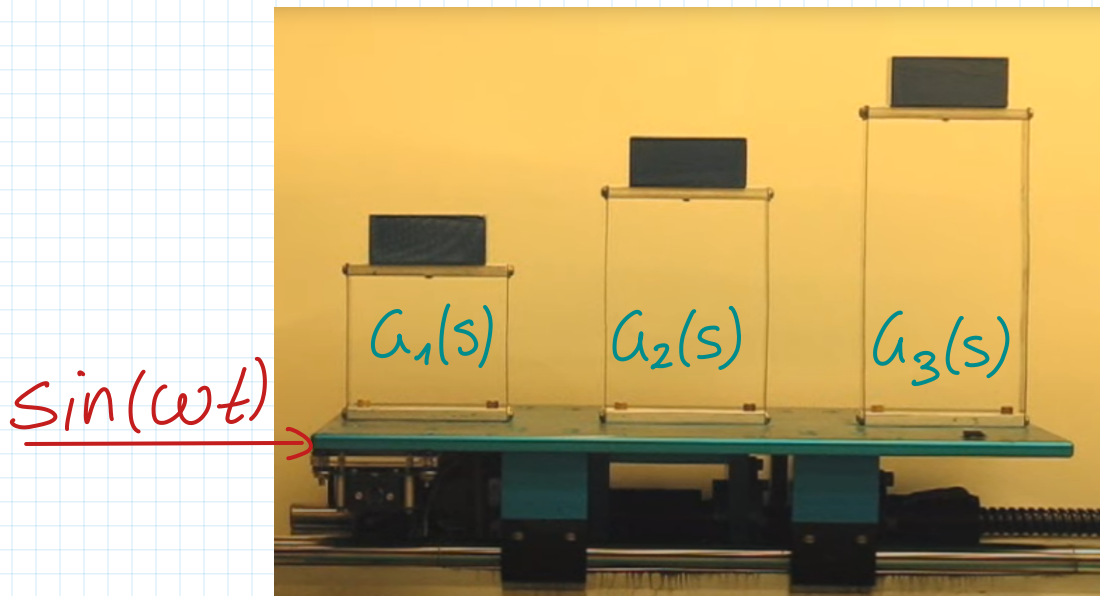
$\Rightarrow$ Plotting this for different s we can find:



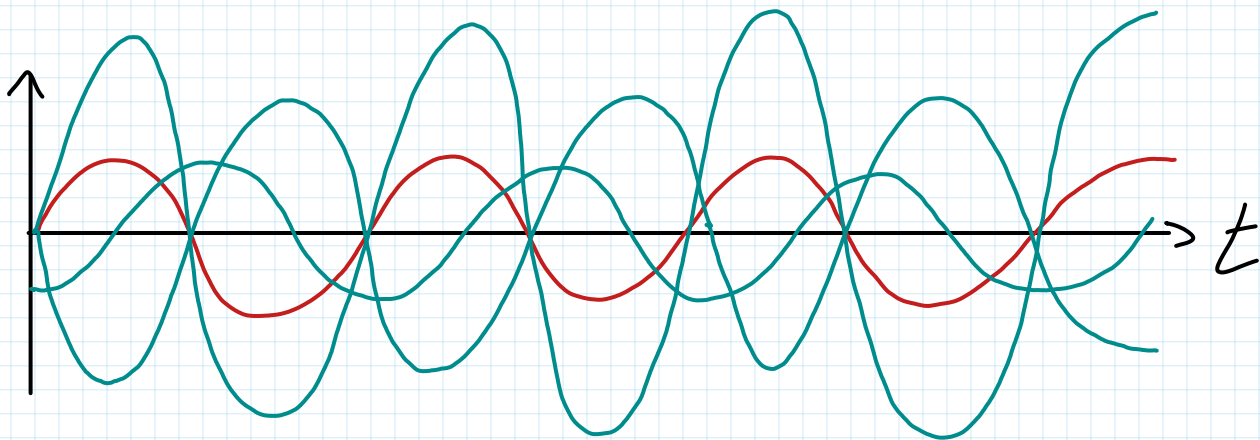
$$\begin{aligned} \Rightarrow \text{We can write } y_{ss}(t) &= \frac{1}{2} \underline{M e^{j\varphi}} e^{j\omega t} + \frac{1}{2} \underline{M e^{-j\varphi}} e^{-j\omega t} \\ &= M \cdot \frac{1}{2} (e^{j(\omega t + \varphi)} + e^{-j(\omega t + \varphi)}) \end{aligned}$$

$$\Rightarrow \text{which is just } y_{ss}(t) = \text{_____} !!!$$

Intuition



$\Rightarrow$ For the same s , these systems have different responses.



- Inputs experience a change in
 - Amplitude: $|G(s_i)|$
 - Phase: $\angle G(s_i)$,
 depending on input frequency.

$\Rightarrow$ Output frequency does not change!!

Now we have fully characterized systems in frequency domain right? No...

• Take another look at the formula for the transfer function $G(s)$...

$$G(s) = C(sI - A)^{-1}B + D$$

$\Rightarrow$ Notice how we said a LTI-system is fully characterized by the following equation:

$$y(t) = \underbrace{C e^{At}}_{\text{homogeneous response}} x(0) + \underbrace{\int_0^t C e^{A(t-\tau)} B u(\tau) d\tau}_{\text{forced response}} + \underline{D}$$

$\Rightarrow$ All parameters included in TF formula, except:

$\Rightarrow$ TF's only show the forced response!

So far so good?

Let's dive in-depth into TF-structure

Transfer Functions: Part II

- Standard form:

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d$$

- Root-Locus form:

$$G(s) = \frac{k_H \cdot (s - z_1)(s - z_2) \dots (s - z_m)}{s^q (s - p_1)(s - p_2) \dots (s - p_{n-q})}$$

- Bode form:

$$G(s) = \frac{k_{\text{Bode}}}{s^q} \cdot \frac{\left(\frac{s}{-z_1} + 1\right)\left(\frac{s}{-z_2} + 1\right) \dots \left(\frac{s}{-z_m} + 1\right)}{\left(\frac{s}{-p_1} + 1\right)\left(\frac{s}{-p_2} + 1\right) \dots \left(\frac{s}{-p_{n-q}} + 1\right)}$$

- The scalar values are called of $G(s)$
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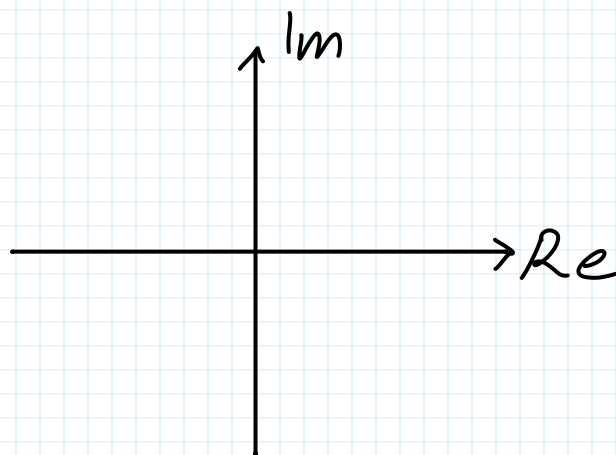
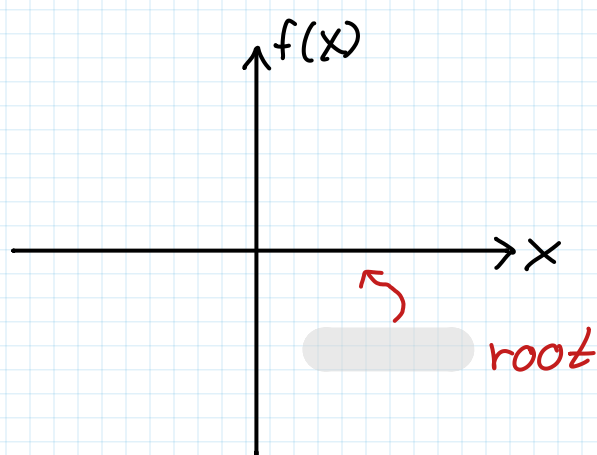
Def: Poles $\Rightarrow$ roots of the denominator of $G(s)$
 Zeros $\Rightarrow$ roots of the numerator of $G(s)$

$$G(s) = \frac{N(s)}{D(s)} \Rightarrow \begin{array}{ll} z_i: & \text{s.t. } \text{ \\ p_i: & \text{s.t. } \text{ \end{array}$$

- Fundamental theorem of algebra:

of polynomial $\hat{=}$ (complex) roots

Ex: $f(x) = x^3 - x^2 - 2 = 0 \Rightarrow$

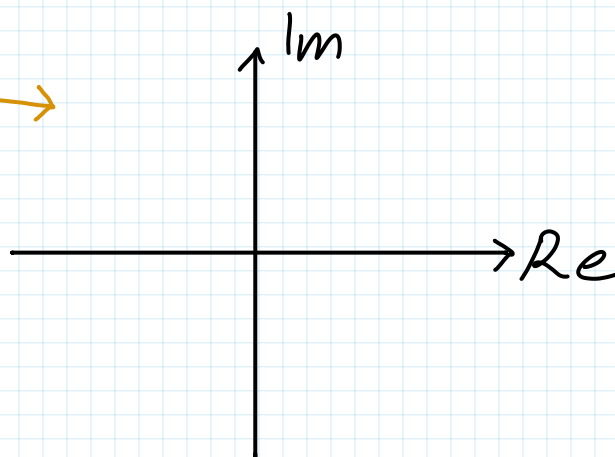


$\Rightarrow$ The location of poles & zeros are the defining characteristic of an LTI-system.

- Graphical notation (given root-locus form)

$$G(s) = \frac{k \cdot \textcolor{blue}{(s+2)} \cdot \textcolor{blue}{(s-1)}}{\textcolor{red}{s} \cdot \textcolor{red}{(s+1+j)} \cdot \textcolor{red}{(s+1-j)}}$$

$\times$: poles, $\circ$: zeros



Poles

Recall: $G(s) = C(sI - A)^{-1}B + D$

From linalg, we had that $A^{-1} =$

$$\Rightarrow G(s) = \frac{1}{\det(sI - A)} \cdot \overbrace{C \cdot \text{adj}(sI - A)B + D} = \frac{N(s)}{\det(sI - A)}$$

$\Rightarrow$ Setting $\det(sI - A) = 0$ is how we find _____!

_____ of $G(s) \hat{=} \text{Eigenvalues of}$ _____

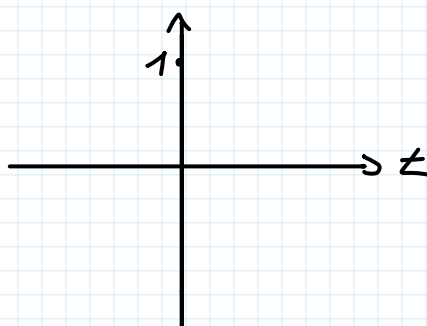
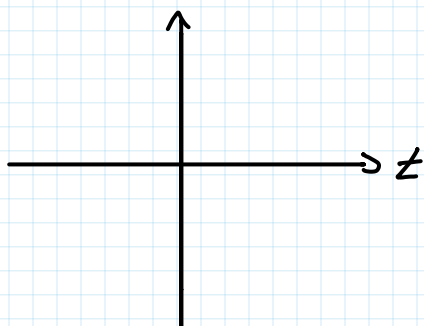
Position of poles will tell us about _____, _____ behaviour and _____ behavior!

To visualize the effect of the position of poles and zeros, we will use two fundamental input signals: $\Rightarrow$ _____ - signals

Test - Signals (some examples)

• _____: $\delta(t) = \begin{cases} 1 & t=0 \\ 0 & t \neq 0 \end{cases}$

• _____ function: $h(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$
(_____ function)



$\Rightarrow$ Why are they relevant? Let's look at $\delta(t)$:

The reaction of a system to a $\delta(t)$ input is called impulse-response:

$$Y(s) = G(s)U(s) =$$

$$\Rightarrow Y(s) = G(s) \quad |$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\{G(s)\}$$

How do we find this?

$x(t)$	$X(s)$
impulse: $\delta(t)$	1
step: $h(t)$	$\frac{1}{s}$
$h(t) \cdot e^{at}$	$\frac{1}{s-a}$
$h(t) \cdot \sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$h(t) \cdot \cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$

• expansion

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0}$$

$$G(s) = \frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} + \dots + \frac{r_n}{s-p_n}$$

r_i :

Of these terms we can easily find $\mathcal{L}^{-1}\{ \}$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{ \frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} + \dots + \frac{r_n}{s-p_n} \right\} \quad | \text{ Laplace is}$$

$$\Rightarrow y(t) =$$

Now we can think of the effect of different poles.

Recall: $s = \sigma + j\omega$

• $e^{\sigma t}$, $\sigma < 0$ • $e^{\sigma t}$, $\sigma > 0$ • $e^{j\omega t} \pm e^{-j\omega t}$



$\Rightarrow$ Matlab

Zeros

To understand the effect of zeros, consider this:

$$u(t) \rightarrow \boxed{\frac{d}{dt}} \rightarrow y(t)$$

$$y(t) = \frac{d}{dt} u(t)$$

$\Downarrow$

Recall this property:

$$\boxed{\mathcal{L}\{f^{(n)}(t)\} = s^n F(s)}$$

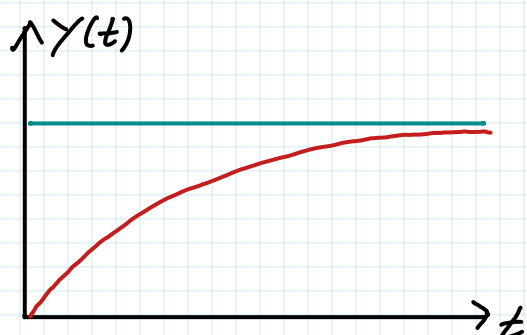
$$\left. \begin{array}{l} Y(s) = \\ \text{New } \omega \text{ in TF!} \end{array} \right\}$$

$\Rightarrow$ Zeros have some connection to derivatives...

How does adding zeros to a system affect their responses?

• Let's try and see this with step responses

- Assume some **system** $G(s)$ without zeros and as input, apply a **step** $h(t)$:



=> How would this change if we added a zero?

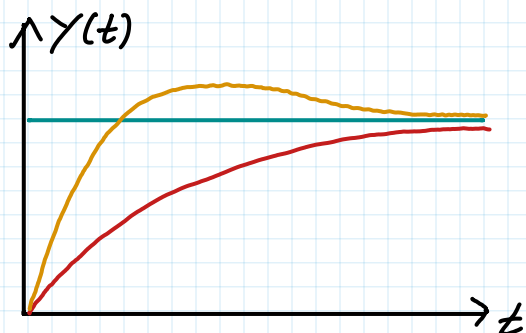
$$\tilde{G}(s) =$$

=

=> The "s" acts as a derivative so the output would look like:

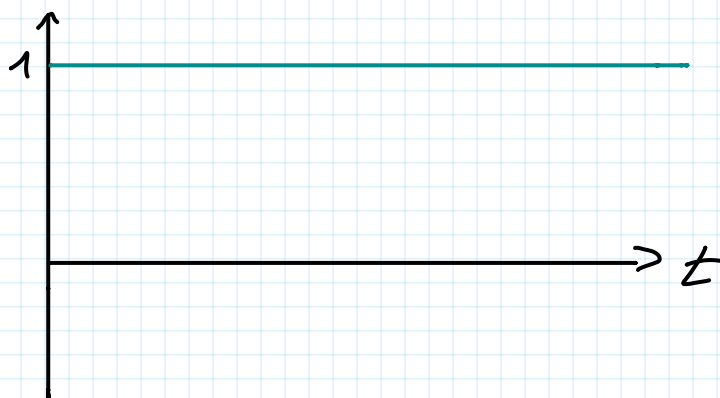
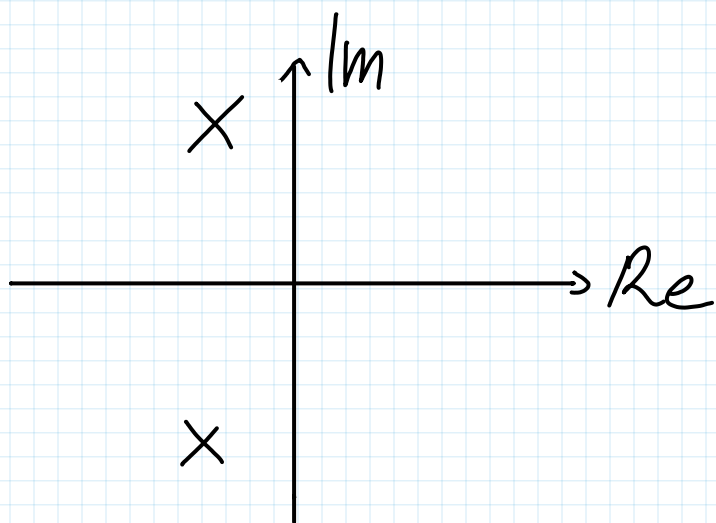
$$\tilde{y}(t) =$$

We add on top a derivative of $y(t)$!



The zero has effect! It "knows in advance" that it is off from the goal.

- Placement of zeros



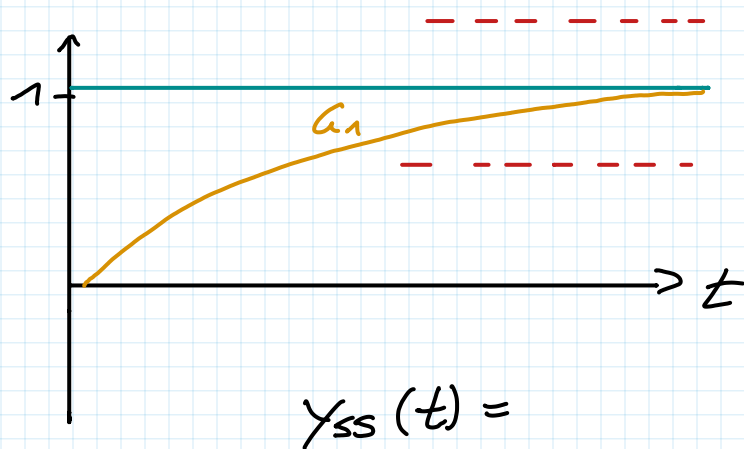
- Zeros in left-half plane: zeros
- Zeros in right-half plane: - minimum phase zeros

↳ Think of how the location of z affects

$$\ddot{y}(t) = -\frac{1}{2}\dot{y}(t) + y(t)$$

- The step response is a really useful tool to analyze systems.

⇒ One thing that we have glossed over...



- A system doesn't necessarily converge to the input value!

- So how could we quantify this ?

 Theorem

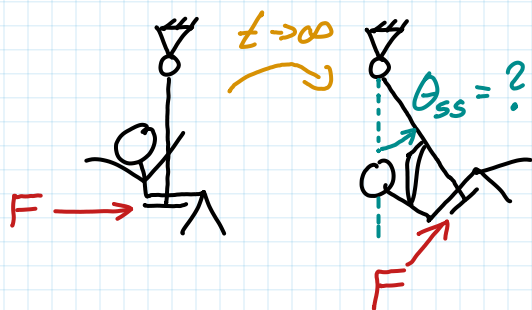
$$\lim_{t \rightarrow \infty} y(t) =$$

- Step function: $y_{ss} =$

$$y_{ss} = \lim_{s \rightarrow 0} = \lim_{s \rightarrow 0}$$

$x(t)$	$X(s)$
impulse: $\delta(t)$	1
step: $h(t)$	$\frac{1}{s}$
$h(t) \cdot e^{at}$	$\frac{1}{s-a}$
$h(t) \cdot \sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$h(t) \cdot \cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$

Example: Step force applied to swing



$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ g/l & -\frac{c}{ml^2} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml} \end{bmatrix} u(t)$$

$$y(t) = [1 \ 0] x(t)$$

$$F(t) = \begin{cases} 0, & t < 0 \\ 1, & \text{else} \end{cases}$$

• We could now solve this as

$$\theta_{ss} = \lim_{t \rightarrow \infty} \left(C e^{At} x_0 + \int_0^t C e^{A(t-\tau)} B d\tau \right)$$

Good luck....

Better idea: transform to frequency domain!

$$G(s) = C(sI - A)^{-1}B + D = \frac{4}{s^2 + s + 2}$$

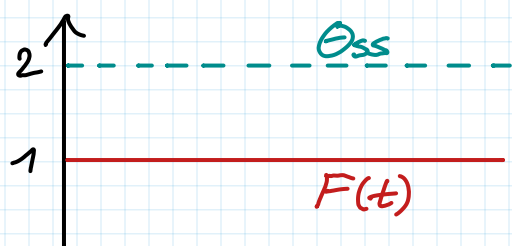
Random params

Then use final value theorem

$$\theta_{ss} = \lim_{s \rightarrow 0} s G(s) U(s), \quad U(s) = H(s) = \frac{1}{s} \quad (\text{step-function})$$

$$= \lim_{s \rightarrow 0} \frac{4}{s^2 + s + 2} =$$

(Degrees, radians,...)

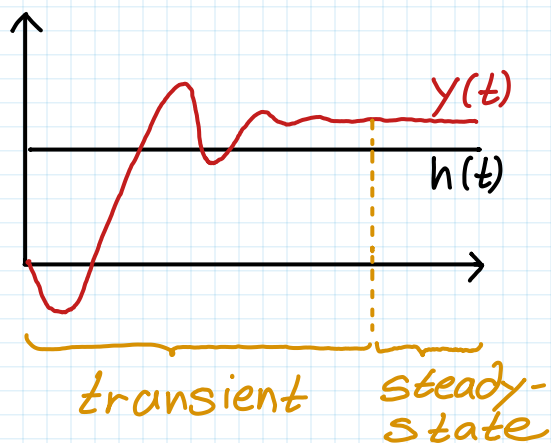


This factor has a name..

$$\Rightarrow G(s \rightarrow 0) =$$

In summary:

=> If you have a system $G(s)$ and want to know how it responds to a step/impulse

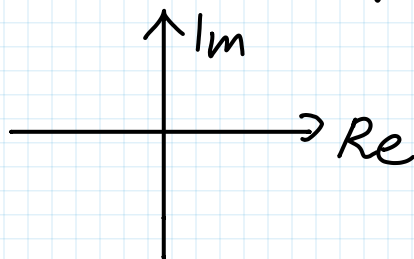


Transient:
Steady-state:

Pole-zero cancellation

• Consider a system with a zero near a pole

$$G(s) = \frac{s+1+\epsilon}{(s+1)(s+2)}$$



what happens if ?

$$G(s) = \frac{s+1}{(s+1)(s+2)} \quad \left\{ \begin{array}{l} \text{The pole and the zero seem} \\ \text{to cancel!} \end{array} \right.$$

But they don't!!!

The internal dynamics remain, we just don't see them anymore in the output! Dangerous if .

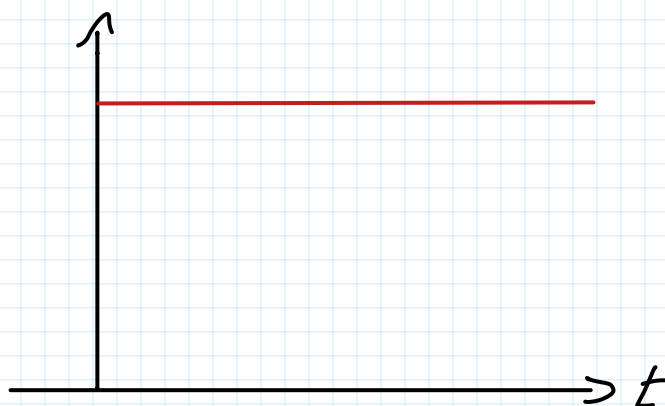
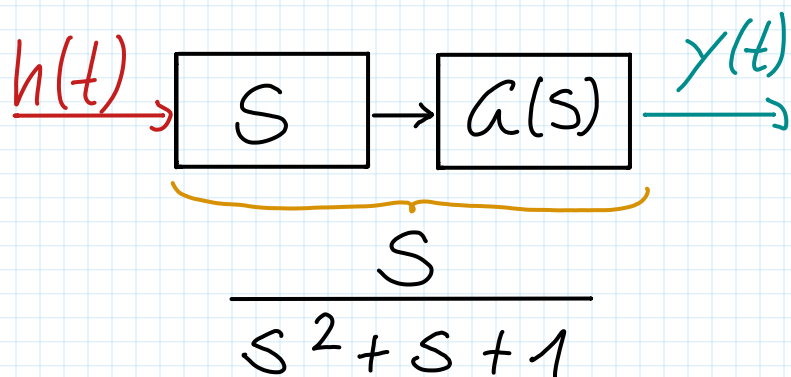
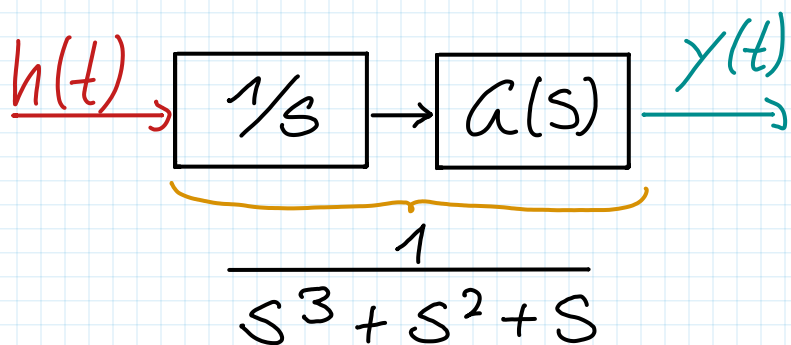
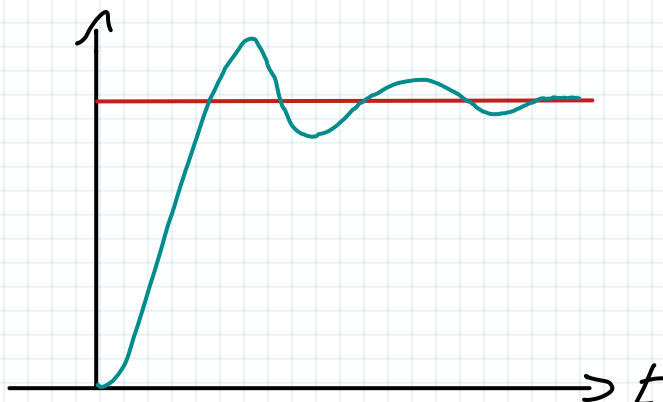
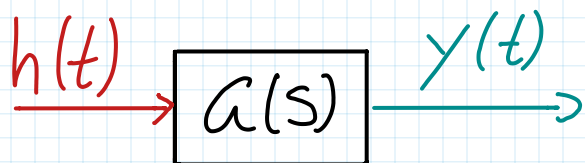
Final note

Poles at $p=0$: Integrators

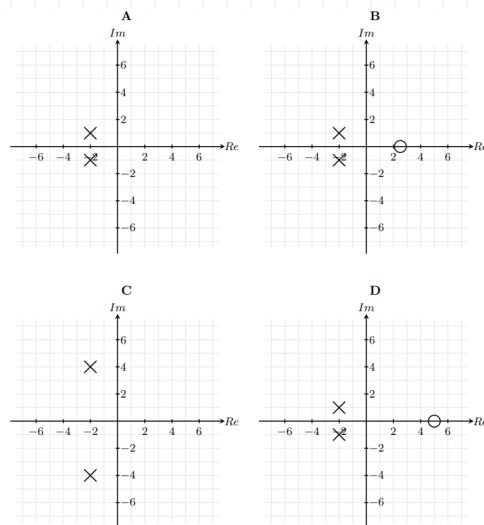
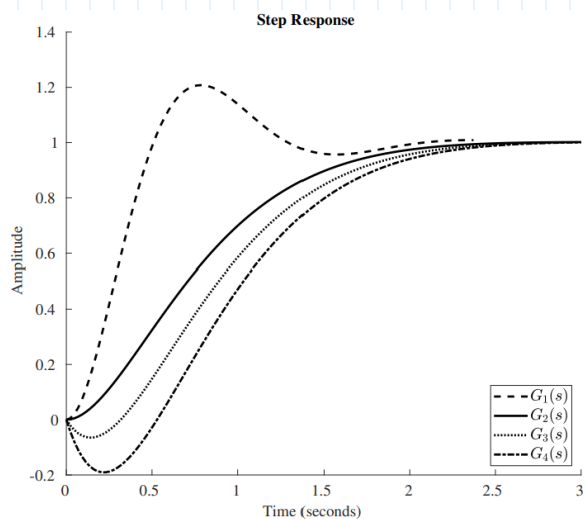
Zeros at $z=0$: Differentiators

$\Rightarrow$ We will use this in controller design

$$G(s) = \frac{1}{s^2 + s + 1}$$



Q1) Match poles/zeros to step response



Q2)

Problem: Consider a linear time-invariant system with transfer function $G(s)$ that is given by a root-locus gain $k_{r1} = 1$ and by two poles and two zeros as shown in Figure 7.

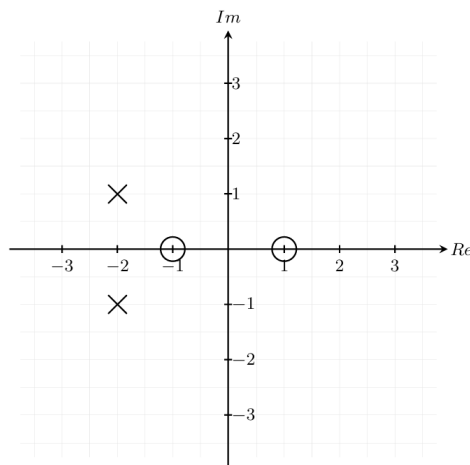


Figure 7: Poles (x) and zeros (o) of the LTI system $G(s)$.

Q19 (1 Points) Mark the correct answer.

What is the magnitude $|G(j)|$ of $G(s = j)$?

- ☐ A $|G(j)| = \frac{1}{2\sqrt{2}}$
☐ B $|G(j)| = \frac{1}{4}$
☐ C $|G(j)| = \frac{1}{2}$
☐ D $|G(j)| = \frac{1}{\sqrt{2}}$

Q20 (1 Points) Mark the correct answer.

What is the phase $\angle G(j)$ of $G(s = j)$?

- ☐ A $\angle G(j) = 45^\circ$
☐ B $\angle G(j) = 0^\circ$
☐ C $\angle G(j) = -45^\circ$
☐ D $\angle G(j) = 135^\circ$

Solution:

Poles:

Zeros:

$$G(s) = \underline{\hspace{2cm}}$$

$$G(s) = \frac{(s-1)(s+1)}{2(2+2j)} = \frac{-2}{4+4j} = -\frac{1}{2+2j}$$

$$\cdot |G(s=j)| = \frac{|-1|}{|2+2j|} = \quad = \quad = \quad //$$

$$\cdot \angle G(s=j) = \angle -\frac{1}{2+2j} =$$

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